

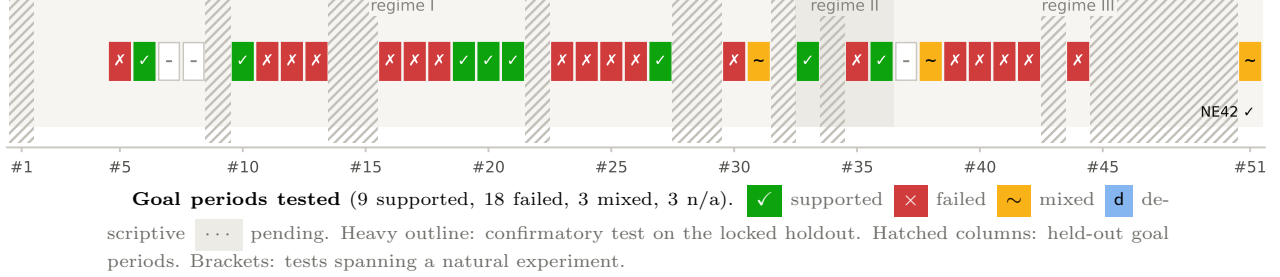
H41: Read-out gating gives the swarm a light cone

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 56% Faithfulness 2/5 Usefulness 2/5

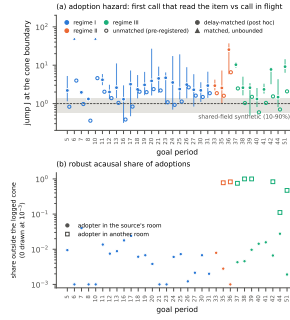
Bottom line. Items obey a logged light cone in rooms; cross-room leaks follow shared projects, with only a small, fragile ordered-read excess ($\Lambda = 1.45$).

level: hypothesis · pipeline checks 71% · updated 2026-10-05



WHAT HAPPENED

H41 asks whether a novel item (a link, number, name or rare word) can reach an agent before that agent's model calls could read it. The test traces 134k novel marker items (45k adoptions) through the DQ1 context ledger in 32 periods. Within a room the logged light cone holds: adoptions outside it are 0–4% per period (median 0.6%). The gating jump at the first reading call passes in 10–12/32 periods as pre-registered and in 22–24/32 when matched for time since the item appeared (post hoc). Across rooms the logged cone is a cage (75–100% of cross-room adoptions outside it in 7/8 two-room periods; the NE42 merge raises cross-group pickup 80–860×). Round 2 time-stamps the leaks with file-level git history and URL fetches. Cross-room adopters read the source's fresh artifacts far more than same-moment non-adopters, but as much after their own use as before: the ordered excess is only $\Lambda = 1.45$ [1.09, 2.26], and it rests on two periods. Pooled with period strata, regime-III numbers gate ($J_{mh,D} = 3.2$); only #40 co-generates them.



Observables. (a) Jump J per period. Open circles: the pre-registered unmatched estimate. It is often ≤ 1 , because the adoption hazard decays with time since t_0 and in-flight calls all sit near t_0 . Filled: delay-matched (post hoc). Gray: shared-field synthetic. (b) Share of adoptions outside the logged cone, for adopters in the source's room (dots, $\leq 4\%$) and in another room (squares; #51's hopping rooms 0.47).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Model 02 gives asynchronous kinetic updates, and model 03 gives contagion on a temporal graph. Agents update only at their model calls. A call reads the room messages posted since the previous call (the ledger). An item that source s posts at t_0 can reach agent i only along a time-respecting read-out path. K_i is i 's entry call into this logged cone. The adoption hazard per talk call is a shared field ε , plus unlogged channels η , plus q once i has read the item. J compares the hazard at the first entry call with the call in flight at t_0 .

$$T_s = t_0, \quad T_i = \min\{t_c^{\text{call}} : c \text{ reads } m, t_m \geq T_{\text{src}(m)}\}$$

$$\lambda_i(c) = \varepsilon(t_c) + \eta_i(c) + q\mathbb{I}[c \geq K_i^{\text{item}}]$$

$$\text{acausal: } c_{\text{use}} < K_i, \quad J = \frac{h(c = K_i)}{h(c = K_i - 1)} \Big|_{\text{same } t-t_0}$$

$$v = 1/\text{median}(\text{calls from exposure to use})$$

Cross-room leaks follow project membership. In #51 a cross-room adopter is almost always an agent that already reads the source's repos (75% vs 3%), before and after the use. To predict where an item leaks, look at who works on the source's artifacts, not at who fetched them just before. Fixed rooms still cage chat: merging two groups multiplies cross-group pickup about a hundredfold, and hopping rooms leak about half the time. Inside a room, names and numbers both gate at the reading call (regime III, J_{mh} 4.8 and 3.2). A dashboard-style common stimulus does not explain the in-flight adoptions. Spread runs at one hop per 1.5–5 talk turns, so call cadence is not a lever.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 J_{in} CI > 1 ($\geq 70\%$ II/III, $\geq 50\%$ I)	3/10 II/III, 6–8/19 I; matched (post hoc) 7–8/11, 15/21	fail (post hoc pass)
P2 $A_{rob} \leq 0.05$ (II/III), 0.10 (I)	29/32; the 3 misses are cross-room; within-room max 0.04	mostly
P6 cadence $b \in [0.5, 1.5]$, $b > 0$ (CI) in 3/25	$c > -0.2$	fail
P8 in-flight $\leq$ entry/3	1–2/10 (matched 3–4/10)	fail
NE42 merge $\geq 3 \times$ both ways; cage	$\times 82$, $\times 860$; cross acausal 1.0 / 0.02 / 1.0	yes
G51 hoppers bridge ($\geq 70\%$ in cone)	0.51 of 839 (1b)	no
R2a pooled Λ CI > 1	1.45 [1.09, 2.26]; lost #35 or #51 (post hoc)	yes, fragile
R2b ordered reads above in-cone controls	L 0.78 [0.66, 0.95]	no
R2c pull the main channel	page 0.44 $>$ pull 0.38	no
R3b lag tracks cone hops	synthetic fields also pass (size 3/8)	not scored
R4a start-time RD for numbers	RD invalid (size 4/12)	not scored
R4b common stimulus enriched in flight	0.24 vs 0.31; ψ_D 1.3 [0.5, 3.6]	no
R4c $J_{mh,D}$ without stimulus > 1	3.0 [2.0, 4.9], but $J_{mh,D}$ is 3.2 anyway	yes (moot)

GOAL PERIODS IN DETAIL

G05 $\times$ failed
G06 $\checkmark$ supported
G10 $\checkmark$ supported
G11 $\times$ failed
G12 $\times$ failed
G13 $\times$ failed
G16 $\times$ failed
G17 $\times$ failed
G18 $\times$ 1b: failed, post hoc supported (was supported: J_in l...
G19 $\checkmark$ supported
+23 more periods: see the card.

FAITHFULNESS SCORECARD

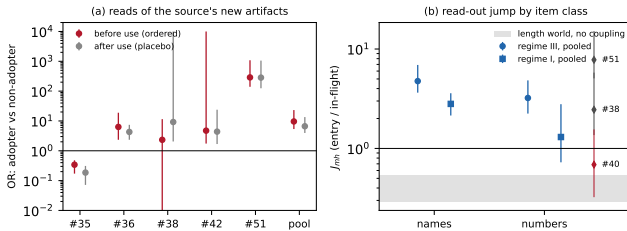
A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The gating result rests on a post-hoc estimator (chosen after the first real period). A synthetic world where talk depends on call length drives it to 0.3–0.5 with no coupling, so real jumps are lower bounds; the start-time RD meant to remove this over-corrects and is not used. Items are hashed markers. Round 2's Λ passes pooled but not with #35 or #51 left out, and in #35 both odds ratios are below 1. Read events come from logged commands and URLs; browser reading without a logged URL is invisible. Visibility follows the ledger's room rule. #40's shared goal confounds NE42. No multiplicity correction across 32 periods. Reserved data are untouched.

ROUND 2



Second figure. Round 2. (a) Odds ratio that a cross-room adopter, vs a same-moment cross-room non-adopter, read an artifact the source had just written: before its own use (red) and in an equal window after it (gray). Their ratio is Λ . In #51 both are ≈ 290 : a shared-project field. (b) Delay-matched jump by class, pooled with period strata; diamonds: the three periods with ≥ 10 in-flight number adoptions. Gray band: J_{mh} in a synthetic world where talk depends on call length and nothing couples.

Find a length-robust in-flight estimator by simulation (size ≤ 0.10) and re-measure co-generation. Model cross-room leaks from the DQ4 co-commit graph and test it out of sample. Trace #40's shared number source from fetch URLs. Pre-register the matched jump and the class split for the reserved data.

- **Essence.** which channels actually move information between agents, and how fast. The logged chat channel is gated at the call scale and bounded by rooms, but a third of fast adoptions and almost all cross-room spread come from elsewhere.
- **Round-3 redirects (newest).**
- **H41-R6** A length-robust jump. Simulate the in-flight vs entry contrast with talk tied to call length at several strengths, and find an estimator with size ≤ 0.10 (e.g.
- **H41-R7** Project membership as the leak variable. Model cross-room adoption as a function of prior co-work on the source's repos (DQ4 co-commit graph) and test whether it predicts leaks out of sample in #51 (and the reserved #47).
- **H41-R8** #40's numbers. The one co-generated period: identify the shared number source (the merged week's common dashboards or APIs) from fetch URLs at the in-flight calls.